

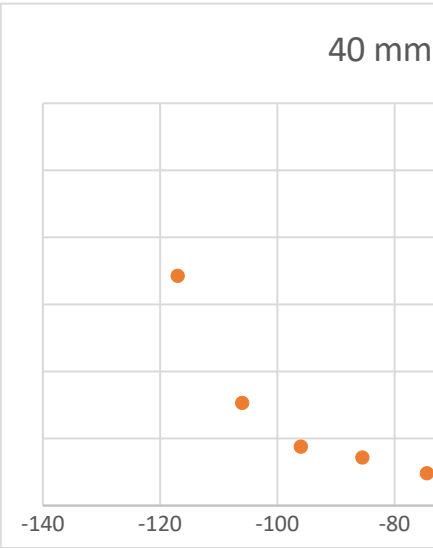
Extensor Test 40mm

Resting muscle length 557
expected max contract 462.31 417.75
tendon 30

	Test #	1	2	3	4	5
	Load (N)	21.395	9.8043	5.4329	1.5627	3.1173
	Knee angle	-117	-106	-96	-66.5	-74.5
Hand	Load cell angle (real ICR)	36.1	29	26.5	32.3	31.2
	Load cell angle (tibia)	32	34.5	31	44.5	35.5
	muscle length	533	529.5	520	524	522
	% contraction	0.0430879713	0.049372	0.066427	0.059245961	0.06283662478
MA	ICR (spline)					
	Tibia origin (matlab)	366.05	366.05	366.05	366.05	366.05
	ICR to muscle	60	65	57	73	62
	dl/dtheta					
	Torque	6.8530172364	3.061645	1.756743	0.428753004	0.96291824495

Moment arm

Torque actual
6.1297215269
2.7479725044
1.5877091528
0.3939683005
0.8799752397
1.3017605699
0.3271082612
5.7663175705
0.9979319577
1.1747393776
5.6669972561
3.0532323371
9.8822570389
5.8339811316



0.05565529623				0.179533			
*				**			
6	7	8	9	10	11	12	13
4.5189	1.1431	20.043	3.3938	4.079	20.853	10.809	33.079
-85.5	-53.5	-53.5	-17	-17	-42.5	-30	-6
27.6	33.6	36.5	36	35.2	39.9	39.7	36.2
33	35.5	35	35	37	40.5	38	34
522	526	470	461	461	462	457	440
0.06283662478	0.05565529623	0.156193895871	0.172352	0.172352	0.170557	0.179533	0.210054
366.05	366.05	366.05	366.05	366.05	366.05	366.05	366.05
67	74	65	70	75	62.5	69	60
1.4332513689	0.35309782369	6.22529896194	1.054105	1.235368	6.042043	3.234337	10.36052
*changed pressure from 100 kPa to 230 kPa							

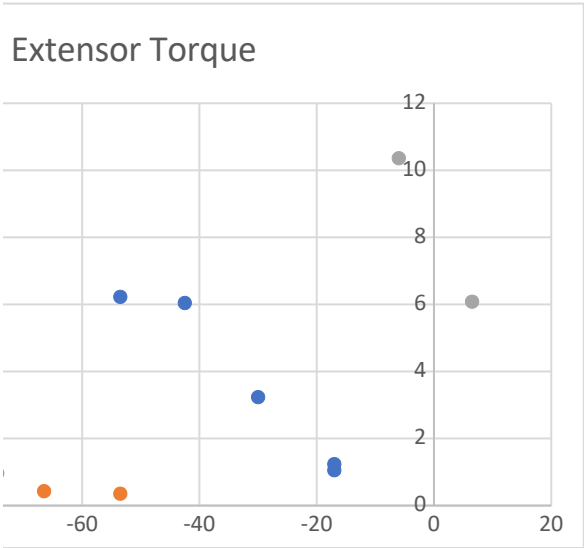
**Increase
Added pul

Vas_Pam insertion wrt arbitrary knee loc

x y z
0.02163 -0.07164 0

Vas_Pam insertion wrt arbitrary knee loc

x y z
0.03776 -0.02481 0



14	15
19.314	
6.5	
32.5	
33.5	
430	
0.228007	
366.05	366.05
64	
6.08095	0

Extensor Test 40mm

Resting muscle length

557

expected max contract
tendon

462.31

90

	Test #	1	2	3	4	5
	Load (N)	84.118	124			
	Knee angle	6	-16			
	Pressure (gauge)	604	530			
Hand	Load cell angle (real ICR)	16.5				
	Load cell angle (tibia)	14.6	14.6	14.6	14.6	14.6
	muscle length	399				
	contraction	0.2836624776				
MA	ICR (spline)					
	Tibia origin (matlab)	240.61	240.61	240.61	240.61	240.61
	ICR to muscle	65				
	dl/dtheta					
	Torque	19.907124595	29.41782	0	0	0
	Human Vas_int Torque	30.557	48			

*Leg assembly started breaking when load cell was i

Vas_Pam insertion wrt arbitrary knee loc

x	y	z	distal
0.02018	-0.11127	0	

Vas_Pam insertion wrt arbitrary knee loc

x	y	z	proximal
0.03619	-0.06395	0	

6	7	8	9	10	11	12	13
14.6	14.6	14.6	14.6	14.6	14.6	14.6	14.6
240.61	240.61	240.61	240.61	240.61	240.61	240.61	240.61
0	0	0	0	0	0	0	0

Moment arm

Torque actual

18.2609

14	15
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14.6	14.6
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240.61	240.61
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0	0
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Extensor Test 20mm

Resting muscle length

expected max contract

actual

tendon length

actual

Hand

MA

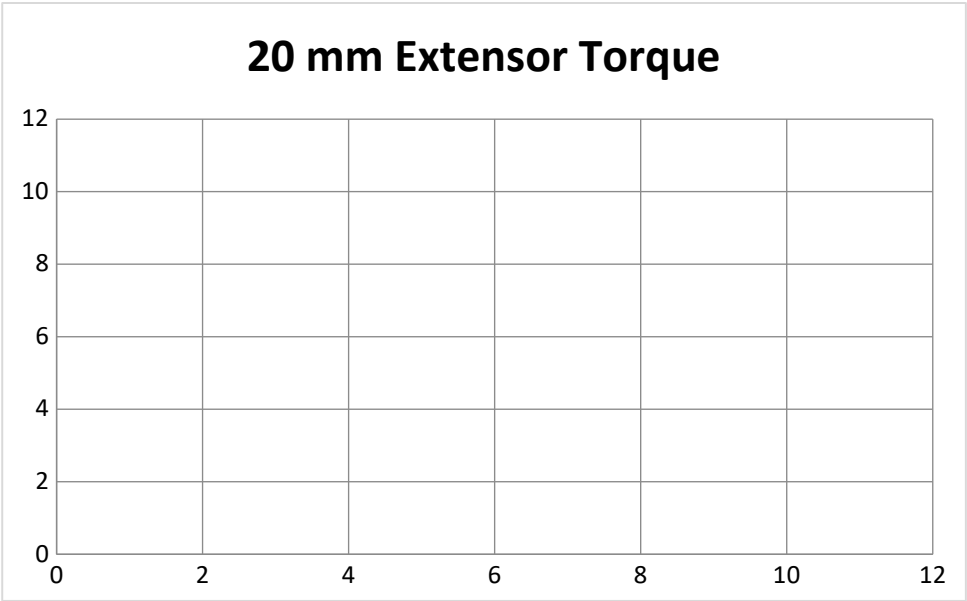
Test #	1	2	3	4	5
Load (N)					
Knee angle					
Pressure (kPa					
Load cell angle (real ICR)					
Load cell angle (tibia)					
muscle length					
ICR (spline)					
Tibia origin (matlab)					
ICR to muscle					
dl/dtheta					
Torque	0	0	0	0	0
Torque actual					

Moment arm

Torque actual

6	7	8	9	10	11	12	13	14

0	0	0	0	0	0	0	0	0



15	16	17	18	19	20	21	22

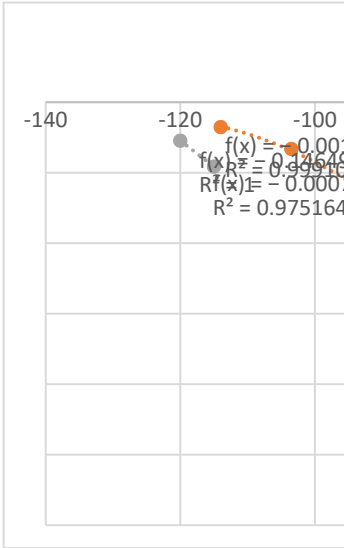
0	0	0	0	0	0	0	0

Flexor Test 20mm

Resting muscle length		423				
expected max contract tendon length		351.09	actual	322	0.238770686	
		17				
	Test #	1	2	3	4	5
	Load (N)	29.225	24.009	20.302	17.246	14.472
	Knee angle	-2	-11.5	-20	-27	-35
	Pressure (kPa	300	300	300	300	300
Hand	Load cell angle (real ICR)	17.4	21.4	22.3	23.1	24.9
	Load cell angle (tibia)	24.5	22	24	25.5	31.5
	muscle length	401	395	389	387	382
MA	ICR (spline)					
	Tibia origin (matlab)	366.43	366.43	366.43	366.43	366.43
	ICR to muscle dl/dtheta	45	47	51	45	40
	Torque	-9.952075229	-8.309776	-6.937209	-5.83121108	-4.6528169083
	Torque actual	-8.843222912	-7.38771	-6.122959	-5.11416856	-4.0057750612

Moment arm

Torque actual



0.126478

*

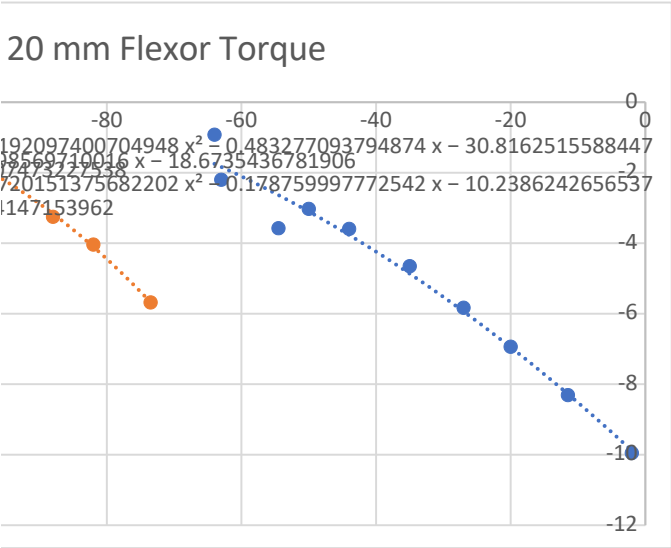
**

6	7	8	9	10	11	12	13
11.127	11.121	9.284	6.8421	2.9257	19.083	14.757	11.612
-44	-54.5	-50	-63	-64	-73.5	-82	-88
300	300	300	300	300	500	500	500
28.4	29.3	29.4	35.1	35.7	34.7	38.3	34.9
31	31.5	30	31.5	33.5	38.5	44.5	43
380	378	374.5	371.5	369.5	354	352.5	350.5
366.43	366.43	366.43	366.43	366.43	366.43	366.43	366.43
37	38	39	36.5	35.5	30.5	20	22

-3.5943174716 -3.5754544525 -3.02655147671 -2.199768 -0.922103 -5.680698 -4.03926 -3.251378
-3.0827195481 -3.0425145123 -2.59384366568 -1.861785 -0.774849 -4.654409 -3.197035 -2.579664

*BPA Started to kink at 7

**Pressure stepped up from 300



14	15	16	17
5.8364	3.1772	7.6226	4.8666
-103.5	-114	-115	-120
500	500	620	620
35.3	42.7	46.4	48.1
54.5	55.5	52	55
345	343.5	338.5	339
366.43	366.43	366.43	366.43
16	4.5	3	10
-1.326358	-0.705989	-1.826208	-1.093715
-0.959099	-0.499572	-1.332995	-0.772724

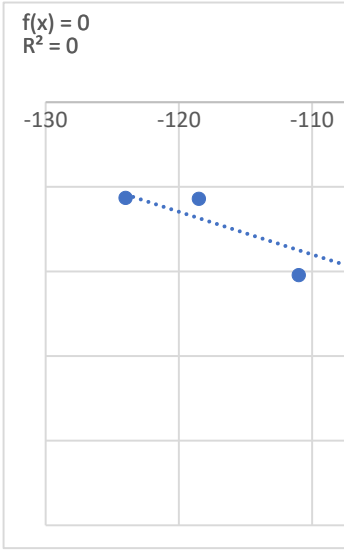
***Pressure stepped up from 500kPa to 620kPa on test 16

Flexor Test 20mm

Resting muscle length		390	"@" 620 kPa		"@" 600 kPa	
expected max contract		292.5	actual	289 0.258974359	290	
tendon length		17 actual				
	Test #	1	2	3	4	5
	Load (N)	18	37	18.5	32	44
	Knee angle	-124	-98.5	-118.5	-111	-90
	Pressure (kPa	610	609	608.75	608	608
Hand	Load cell angle (real ICR)					
	Load cell angle (tibia)	29	31	31	27	28
	muscle length	330	331	328	330	333
MA	ICR (spline)					
	Tibia origin (matlab)	350	350	350	350	350
	ICR to muscle	10	25	15	20	25
	dl/dtheta					
	Torque	-5.654183337	-11.41608	-5.708041	-10.2181478	-13.937767935
	Torque actual					

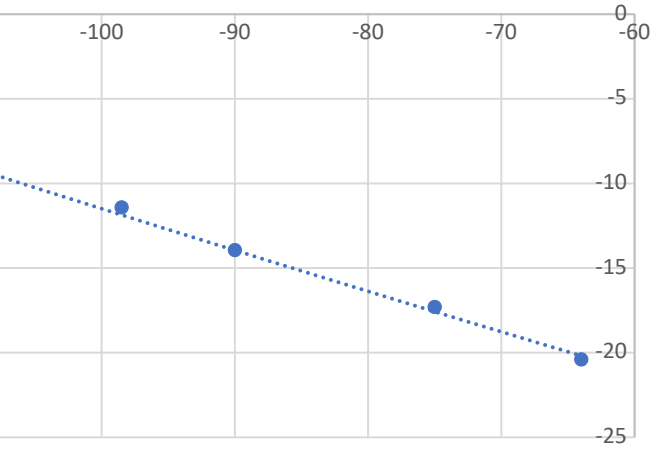
Moment arm

Torque actual



0.25641025641				0.126478			
*				**			
6	7	8	9	10	11	12	13
53	61						
-75	-64						
608	606.5						
24	20						
337							
350	350	350	350	350	350	350	350
25							
-17.298116513	-20.398495907	0	0	0	0	0	0

20 mm Flexor Torque



14

15

16

17

350

350

350

0

0

0

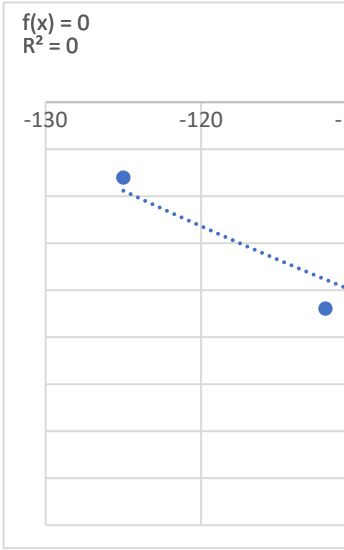
0

Flexor Test 20mm

Resting muscle length		390	"@" 620 kPa		"@" 600 kPa
expected max contract		292.5	actual	289 0.258974359	290
tendon length		25 actual	*		
	Test #	1	2	3	4
	Load (N)	10.6	28.6	36.9	50.45
	Knee angle	-125	-112	-91.5	-75
	Pressure (kPa	608	610	610	608
Hand	Load cell angle (real ICR)				
	Load cell angle (tibia)	33	31.5	30	26.5
MA	muscle length	325	327	332	335
	ICR (spline)				
	Tibia origin (matlab)	350	350	350	350
	ICR to muscle	10	15	25	25
	dl/dtheta				
Torque		-3.207436218	-8.782749	-11.4899	-16.1720262
Torque actual		0			
		*tibia ring broke			

Moment arm

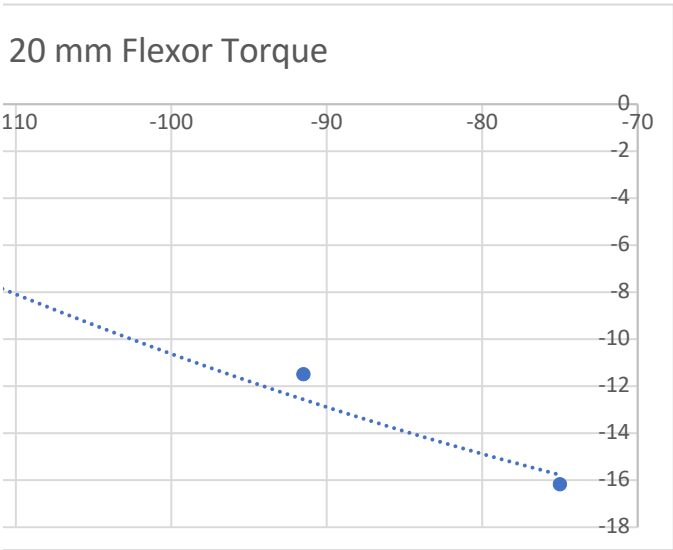
Torque actual



0.25641025641

0.126478

6	7	8	9	10	11	12	13
350	350	350	350	350	350	350	350
0	0	0	0	0	0	0	0



14	15	16	17
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350	350	350	
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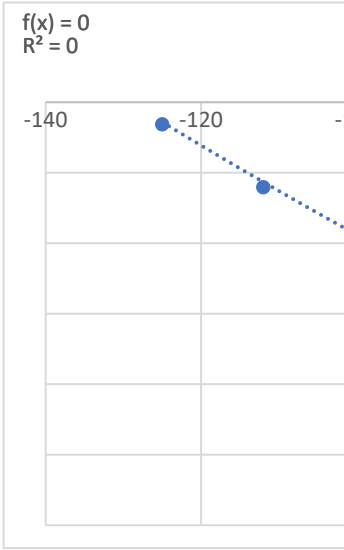
0	0	0	0
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Flexor Test 20mm

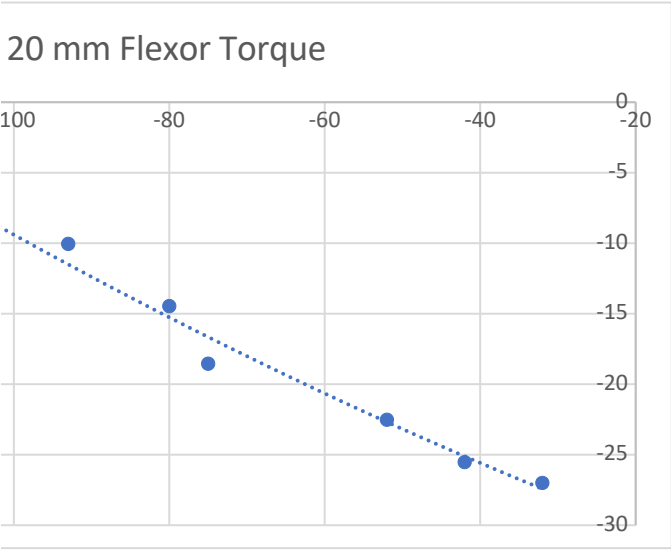
Resting muscle length		420	"@" 620 kPa		"@" 600 kPa	
expected max contract		315	actual	1		
tendon length		15 actual				
	Test #	1	2	3	4	5
	Load (N)	5.3	19.83	31.35	44.78	56.1
	Knee angle	-125	-112	-93	-80	-75
	Pressure (kPa	614	613.57	613.57	618	617
Hand	Load cell angle (real ICR)					
	Load cell angle (tibia)	35	32.5	26.5	25.5	22
	muscle length	330	335	339	345	350
MA	ICR (spline)					
	Tibia origin (matlab)	350	350	350	350	350
	ICR to muscle	20	25	30	37	40
	dl/dtheta					
	Torque	-1.570205678	-6.030537	-10.04942	-14.4621044	-18.54620852
	Torque actual	-1.5952	-6.0761	-10.0058	-14.3527	-18.339

Moment arm

Torque actual



6	7	8	9	10	11	12	13
68.12	74.9	78.14					
-52	-42	-32					
615	618	620					
22	16	12					
355	357	360					
350	350	350	350	350	350	350	350
45	45	45					
-22.519923786	-25.525501817	-26.9994754957	0	0	0	0	0
-22.1908							



14	15	16	17
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350	350	350	
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0	0	0	0
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Flexor Test 20mm

Resting muscle length

415

"@" 620 kPa

"@" 600 kPa

expected max contract

314

actual

1

tendon length

14

actual

	Test #	1	2	3	4	5
	Load (N)	18.5	18.05	32	40.97	44.4
	Knee angle	-125	-114	-98	-83	-75.5
	Pressure (kPa	613	614	614	615	615
Hand	Load cell angle (real ICR)					
	Load cell angle (tibia)	32.5	30	28	26	24.5
	muscle length	334	334	338	343	NaN
MA	ICR (spline)					
	Tibia origin (matlab)	350	350	350	350	350
	ICR to muscle	25	25	35	35	NaN
	dl/dtheta					
	Torque	-5.626068039	-5.620399	-10.13656	-13.1828972	-14.44172671
	Torque actual	-5.6922	-5.6461	-10.119	-13.0959	-14.3096

Moment arm

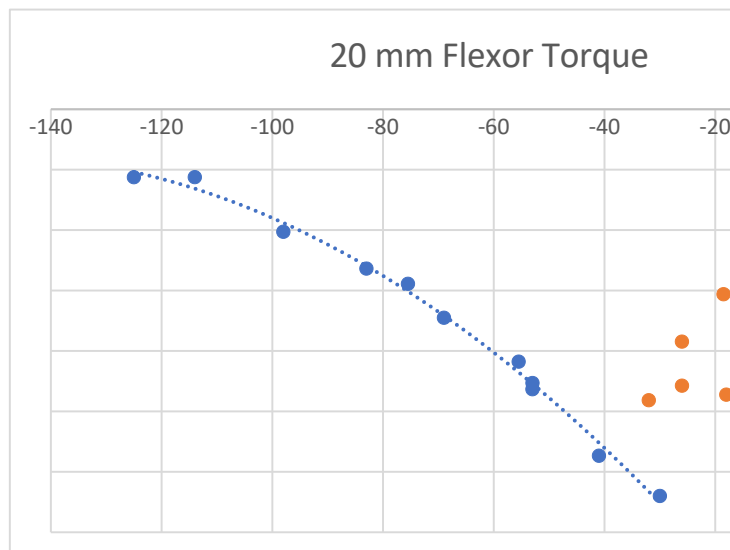
Torque actual

avg pres
616.5454545

*

6	7	8	9	10	11	12	13
50.84	62.45	69.4	91.7	114.1	129.8	78	92.78
-69	-55.5	-53.001	-53.01	-41	-30	-26.01	-26.001
615	618	618	620	620	620	385	451
17	20	24	-5	-7	-12	-12	-12
348	356	356	357	365	368	381	377
350	350	350	255	255	255	255	255
37	40	45	48	45	40	40	40
-17.2525937	-20.883378186	-22.6507412458	-23.16549	-28.66834	-31.99646	-19.22745	-22.87081
-16.9889	-20.5703	-22.3443	-21.6006	-26.7782	-29.8989	-17.9934	-21.352

*Switched to shorter tibia



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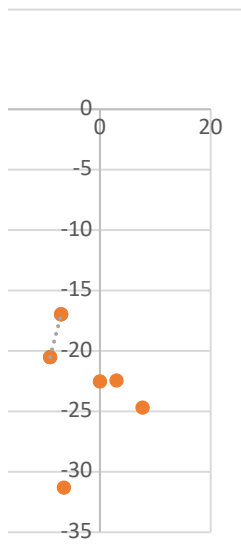
14	15	16	17	18	19	20	21	22	23
63.6	97.28	71.92	84.5	93.9	64.3	70.6	90.15	70	
-18.5	-18	-7	-9	0	3	7.7	-6.5	-32	
296.6	421	281	324.8	325.58	325.58	325	500	560	
-16.5	-15	-19.5	-15	-17	7	5.5	10	13.5	
387	385	396	392	397	398	401	384	367	
255	255	255	255	255	350	350	350	350	350
42	45	46	48	48	42	43	45	38	
-15.30376	-23.61493	-16.96433	-20.51256	-22.52467	-22.44542	-24.68317	-31.30577	-24.07639	0
-14.3414	-22.1444	-15.9752	-19.3052	-21.282	-21.9932	-24.2025	-30.6504	-23.6068	

*Switched back to long tibia

** 22.4N ρ **27.6N pretension

***411mm uninflated

Max extension possible with current knee



23	23	23	23	23
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350	350	350	350	350
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0	0	0	0	0
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Flexor Test 20mm

Resting muscle length

expected max contract

actual

tendon length

actual

Test #	1	2	3	4	5
Load (N)					
Knee angle					
Pressure (kPa					
Hand					
Load cell angle (real ICR)					
Load cell angle (tibia)					
muscle length					
MA					
ICR (spline)					
Tibia origin (matlab)					
ICR to muscle					
dl/dtheta					
Torque	0	0	0	0	0
Torque actual					

Moment arm

Torque actual

6	7	8	9	10	11	12	13	14

0	0	0	0	0	0	0	0	0



15	16	17	18	19	20	21	22

0	0	0	0	0	0	0	0